Project: Worked examples
Title: Psychrometrics
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

The state of moist air (humidity ratio, relative humidity, dew point, wet-bulb, enthalpy, specific volume, density) from dry-bulb and one other property at a stated pressure or altitude, and one air-handling process from that state with its heat and moisture balance: sensible heating or cooling, cooling with dehumidification, steam humidification, adiabatic humidification, or mixing with a second stream.

WHAT IT CALCULATES	RELATION
Saturation vapour pressure at the dry-bulb temperature, over water at or above 0 °C and over ice below	$p_{ws}(t)$ by Hyland–Wexler
Moisture content (humidity ratio) of the air	$W = 0.621945\ p_w/(p - p_w)$
Relative humidity	$\varphi = p_w/p_{ws}$
Specific enthalpy per kilogram of dry air	$h = 1.006\ t + W(2501 + 1.86\ t)$ kJ/kg$_{da}$
Specific volume	$v = 0.287042\ (t + 273.15)(1 + 1.607858\ W)/p$
Dew point and wet-bulb temperatures, by bisection (wet-bulb from the adiabatic-saturation relation)	$p_{ws}(t_{dp}) = p_w$
Sensible heating or cooling at constant moisture content	$Q = \dot{m}\ (h_2 - h_1)$
Cooling with dehumidification, less the heat leaving in the condensate	$Q_{total} = \dot{m}(h_1 - h_2) - \dot{m}_w\ h_w$, with $\dot{m}_w = \dot{m}(W_1 - W_2)$
Sensible and latent split of the coil load, along the path $t_1 \rightarrow t_2$ at W_1 then $W_1 \rightarrow W_2$ at t_2	$Q_{sensible} = \dot{m}(h_1 - h(t_2, W_1))$; $Q_{latent} = \dot{m}(h(t_2, W_1) - h_2)$; $SHR = Q_{sensible}/(Q_{sensible} + Q_{latent})$. With RH_2 blank and t_2 above the entering dew point the coil is dry: $W_2 = W_1$, no condensate, sensible only
Steam humidification	$h_2 = h_1 + (W_2 - W_1)\ h_{steam}$
Adiabatic (spray) humidification, at constant enthalpy	$h_2 = h_1$
Mixing two airstreams, weighted by dry-air mass flow	$W_{mix} = \Sigma\dot{m}\ W/\Sigma\dot{m}$; $h_{mix} = \Sigma\dot{m}\ h/\Sigma\dot{m}$

Basis: Psychrometrics (ideal-gas relations, processes)[1](Ch. 1); Saturation pressure of water and ice[2]; Properties of humid air (principles)[3](Section 1).

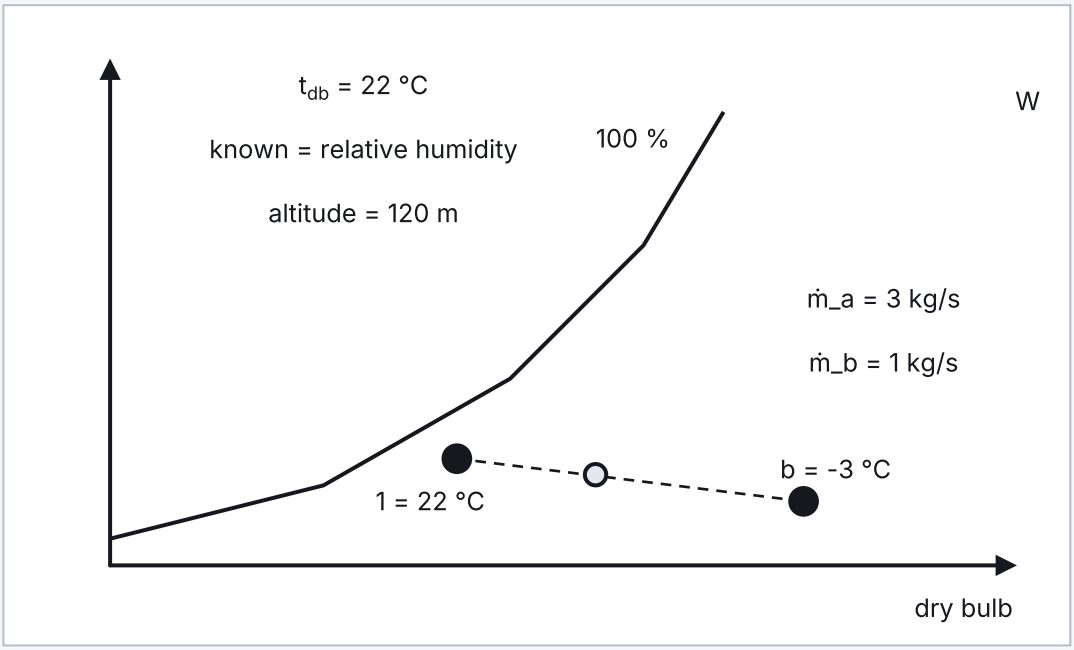


Figure 1 — Adiabatic mixing of two streams: the mixed state lies on the line between them, by mass flow. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
altitude	Altitude (used when pressure is blank)	120	m	drawing: Site survey SRV-01: 120 m AOD	-500 … 5000 m
t_{db}	Dry-bulb temperature	22	°C	spec: Return air 22 °C / 40 %	-40 … 60 °C
known	Second known property	relative humidity		assumption: RH given.	any
RH	Relative humidity	40		spec: Return air 22 °C / 40 %	0 … 100
m_{da}	Dry-air mass flow	3	kg/s	spec: AHU-02 return 3 kg/s	0 … 1000 kg/s
process	Process	adiabatic mixing		assumption: Mixing box.	any
h_{steam}	Steam enthalpy (steam humidification)	2676		default	2500 … 3000
$t_{db,b}$	Second stream dry-bulb (mixing)	-3	°C	spec: CIBSE Guide A winter design −3 °C / 90 %, assumed	-40 … 60 °C
RH_b	Second stream relative humidity (mixing)	90		spec: Winter design condition	0 … 100
$m_{da,b}$	Second stream dry-air mass flow (mixing)	1	kg/s	spec: AHU-02 minimum outdoor air 1 kg/s	0 … 1000 kg/s

3. Assumptions

A1. Moist air is an ideal mixture of ideal gases; the enhancement factor is taken as 1. (ASHRAE Fundamentals Ch. 1 ideal-gas relations.)
A2. Condensate leaves at the coil leaving dry-bulb; steam adds enthalpy h_{steam} at negligible mass of dry air. (Process conventions.)
A3. Adiabatic humidification follows a line of constant enthalpy (close to constant wet-bulb). (Evaporative process idealisation.)
A4. known (Second known property) is taken as relative humidity. (RH given.)
A5. process (Process) is taken as adiabatic mixing. (Mixing box.)

4. Working

[2] p = 101.325 (1 - 2.25577e-5 z)^5.2559 [1](Ch. 1 eq. 3)
= z = 120 m
= 99.892 kPa
Barometric pressure (standard atmosphere)

[6] $p_{ws,1}$ = p_{ws}(t) Hyland–Wexler (entering)
= t = 22 °C
= 2.6448 kPa
Saturation pressure, state 1

[7] $p_{w,1}$ = p W/(0.621945 + W)
= W = 0.0066572
= 1.0579 kPa
Vapour pressure, state 1

[8] $t_{1,out}$ = dry-bulb (entering)
=
= 22 degC
Dry-bulb, state 1

[9] W_1 = 0.621945 p_w/(p - p_w)
= 1.0579 / (99.892 - 1.0579)
= 6.6572
Humidity ratio, state 1 (g/kg dry air)

[10] RH_1 = p_w / p_{ws}
= 1.0579 / 2.6448
= 40
Relative humidity, state 1 (%)

[11] φ_1 = p_w / p_{ws}
= 1.0579 / 2.6448
= 40 %
Relative humidity, state 1, as a percentage quantity (the same value as RH_1; a plain-number output cannot be linked to a percentage input)

[12] $t_{dp,1}$ = p_{ws}(t_{dp}) = p_w, bisection
= p_w = 1.0579 kPa
= 7.7942 degC
Dew point, state 1

[13] $t_{wb,1}$ = adiabatic-saturation relation, bisection
= t = 22, W = 0.0066572
= 13.853 degC
Wet-bulb, state 1

[14] h_1 = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 22 + 0.0066572 × (2501 + 1.86 × 22)
= 39.054
Enthalpy, state 1 (kJ/kg dry air)

[15] v_1 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p
= p = 99.892 kPa
= 0.8572 m³/kg
Specific volume, state 1 (per kg dry air)
[16] ρ_1 = (1 + W)/v
=
= 1.1744 kg/m³
Moist-air density, state 1
[22] $p_{ws,b}$ = p_{ws}(t) Hyland–Wexler (second stream)
= t = -3 °C
= 0.47606 kPa
Saturation pressure, state b
[23] $p_{w,b}$ = p W/(0.621945 + W)
= W = 0.0026791
= 0.42845 kPa
Vapour pressure, state b
[24] $t_{b,out}$ = dry-bulb (second stream)
=
= -3 degC
Dry-bulb, state b
[25] W_b = 0.621945 p_w/(p − p_w)
= 0.42845 / (99.892 − 0.42845)
= 2.6791
Humidity ratio, state b (g/kg dry air)
[26] RH_b = p_w / p_{ws}
= 0.42845 / 0.47606
= 90
Relative humidity, state b (%)
[27] φ_b = p_w / p_{ws}
= 0.42845 / 0.47606
= 90 %
Relative humidity, state b, as a percentage quantity (the same value as RH_b; a plain-number output cannot be linked to a percentage input)
[28] $t_{dp,b}$ = p_{ws}(t_{dp}) = p_w, bisection
= p_w = 0.42845 kPa
= -4.2454 degC
Dew point, state b
[29] $t_{wb,b}$ = adiabatic-saturation relation, bisection
= t = -3, W = 0.0026791
= -3.4949 degC
Wet-bulb, state b
[30] h_b = 1.006 t + W (2501 + 1.86 t)
= 1.006 × -3 + 0.0026791 × (2501 + 1.86 × -3)
= 3.6675
Enthalpy, state b (kJ/kg dry air)
[31] v_b = 0.287042 (t + 273.15)(1 + 1.607858 W)/p
= p = 99.892 kPa
= 0.77963 m³/kg
Specific volume, state b (per kg dry air)
[32] ρ_b = (1 + W)/v
=
= 1.2861 kg/m³
Moist-air density, state b

[33] $m_{da,3}$ = ṁ_a + ṁ_b
= 3 + 1
= 4 kg/s
Mixed dry-air mass flow
[34] $p_{ws,2}$ = p_{ws}(t) Hyland-Wexler (mixed)
= t = 15.784 °C
= 1.7935 kPa
Saturation pressure, state 2
[35] $p_{w,2}$ = p W/(0.621945 + W)
= W = 0.0056627
= 0.90129 kPa
Vapour pressure, state 2
[36] $t_{2,out}$ = dry-bulb (mixed)
=
= 15.784 degC
Dry-bulb, state 2
[37] W_2 = 0.621945 p_w/(p - p_w)
= 0.90129 / (99.892 - 0.90129)
= 5.6627
Humidity ratio, state 2 (g/kg dry air)
[38] RH_2 = p_w / p_{ws}
= 0.90129 / 1.7935
= 50.253
Relative humidity, state 2 (%)
[39] φ_2 = p_w / p_{ws}
= 0.90129 / 1.7935
= 50.253 %
Relative humidity, state 2, as a percentage quantity (the same value as RH_2; a plain-number output cannot be linked to a percentage input)
[40] $t_{dp,2}$ = p_{ws}(t_{dp}) = p_w, bisection
= p_w = 0.90129 kPa
= 5.4665 degC
Dew point, state 2
[41] $t_{wb,2}$ = adiabatic-saturation relation, bisection
= t = 15.784, W = 0.0056627
= 10.314 degC
Wet-bulb, state 2
[42] h_2 = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 15.784 + 0.0056627 × (2501 + 1.86 × 15.784)
= 30.207
Enthalpy, state 2 (kJ/kg dry air)
[43] v_2 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p
= p = 99.892 kPa
= 0.83782 m³/kg
Specific volume, state 2 (per kg dry air)
[44] ρ_2 = (1 + W)/v
=
= 1.2003 kg/m³
Moist-air density, state 2

5. Results

SYMBOL	VALUE	DESCRIPTION
$t_{1,out}$	22 °C	Dry-bulb 1

SYMBOL	VALUE	DESCRIPTION
$p_{ws,1}$	2.645 kPa	p_{ws} 1
$p_{w,1}$	1.058 kPa	p_w 1
W_1	6.6572	W 1 (g/kg)
RH_1	40	RH 1 (%)
φ_1	40 %	φ 1 (RH as a % quantity, for linking)
$t_{dp,1}$	7.794 °C	Dew point 1
$t_{wb,1}$	13.85 °C	Wet-bulb 1
h_1	39.054	h 1 (kJ/kg)
v_1	0.8572 m³/kg	v 1
ρ_1	1.174 kg/m³	ρ 1
$t_{b,out}$	-3 °C	Dry-bulb b
$p_{ws,b}$	0.4761 kPa	p_{ws} b
$p_{w,b}$	0.4285 kPa	p_w b
W_b	2.6791	W b (g/kg)
RH_b	90	RH b (%)
φ_b	90 %	φ b (RH as a % quantity, for linking)
$t_{dp,b}$	-4.245 °C	Dew point b
$t_{wb,b}$	-3.495 °C	Wet-bulb b
h_b	3.6675	h b (kJ/kg)
v_b	0.7796 m³/kg	v b
ρ_b	1.286 kg/m³	ρ b
$t_{2,out}$	15.78 °C	Dry-bulb 2
$p_{ws,2}$	1.794 kPa	p_{ws} 2
$p_{w,2}$	0.9013 kPa	p_w 2
W_2	5.6627	W 2 (g/kg)
RH_2	50.253	RH 2 (%)
φ_2	50.25 %	φ 2 (RH as a % quantity, for linking)
$t_{dp,2}$	5.467 °C	Dew point 2
$t_{wb,2}$	10.31 °C	Wet-bulb 2
h_2	30.207	h 2 (kJ/kg)
v_2	0.8378 m³/kg	v 2
ρ_2	1.200 kg/m³	ρ 2
$m_{da,3}$	4 kg/s	Mixed mass flow

6. Checks (governing first)

None.

7. Applicability

None.

Defaults used (inputs left empty):
h_{steam}: not entered, so the default of 2676 is used.

8. Verification

```
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Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. American Society of Heating, Refrigerating and Air-Conditioning Engineers. ASHRAE Handbook—Fundamentals. ASHRAE.
2. Hyland RW, Wexler A. Formulations for the thermodynamic properties of the saturated phases of H2O from 173.15 K to 473.15 K. ASHRAE Trans. 1983;89(2A):500-519.
3. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.